ITSM RAG Testing Guide

Sample Knowledge Document for AI Knowledge Assistant

Purpose: This small document is designed specifically to test Retrieval-Augmented Generation (RAG) in an IT Service Management dashboard. The content is intentionally structured with clear definitions, procedures, priorities, and examples so that questions can be tested against known answers.

1. Incident Management

Incident: An incident is an unplanned interruption to an IT service, or a reduction in the quality of an IT service. The primary objective of Incident Management is to restore normal service operation as quickly as possible and minimize the impact on users and the business.

Example: If employees cannot log in to the corporate ERP system, the event is treated as an incident because an existing service is unavailable to users.

Incident Management Objectives

The main objectives are to restore service quickly, reduce business impact, maintain effective communication with stakeholders, record incident information accurately, and identify incidents that require escalation or further investigation.

2. Incident Lifecycle

A typical incident lifecycle contains the following activities: identification, logging, categorization, prioritization, initial diagnosis, investigation and diagnosis, resolution and recovery, and closure.

Incident Prioritization

3. Impact and Urgency

Impact measures how many users, services, or business processes are affected. **Urgency** measures how quickly the incident needs to be resolved based on business criticality and time sensitivity.

Priority	Typical Situation	Example
P1 - Critical	Very high impact and high urgency	Major business service unavailable
P2 - High	Significant impact or urgent business need	Important application unavailable for a department
P3 - Medium	Limited impact with normal urgency	Single user unable to access a non-critical service
P4 - Low	Low impact and low urgency	Minor service issue or information request

4. Incident Escalation

Functional escalation transfers an incident to a specialist support team when the current team does not have the required technical knowledge. Hierarchical escalation involves informing management or business stakeholders when the incident has high business impact, requires urgent decisions, or risks missing an important deadline.

Example: If a payroll application is unavailable shortly before payroll processing, the incident may require both technical escalation to the application support team and management escalation because of the business deadline.

Problem Management

5. What is a Problem?

A problem is a cause, or potential cause, of one or more incidents. Problem Management focuses on reducing the likelihood and impact of incidents by identifying underlying causes and managing workarounds and known errors.

6. Incident Management vs Problem Management

Area	Incident Management	Problem Management
Primary focus	Restore service quickly	Find and manage underlying causes
Main question	How can we restore the service?	Why did this happen?
Typical output	Restored service and incident closure	Root cause, workaround, known error, or corrective action
Time horizon	Immediate or short term	Medium or long term

7. Root Cause Analysis

Root Cause Analysis (RCA) is a structured approach used to identify the fundamental cause of an incident or problem rather than only treating its symptoms. Common RCA techniques include the 5 Whys, Fishbone Diagram, Pareto Analysis, and Fault Tree Analysis.

Example: If a server repeatedly becomes unavailable because its disk reaches 100% utilization, the problem investigation should determine why disk usage grows, whether monitoring is adequate, and what permanent corrective action is required.

Change Management and Service Improvement

8. Change Management

A change is the addition, modification, or removal of anything that could have a direct or indirect effect on services. Change Management aims to maximize the number of successful changes while assessing risk, coordinating implementation, and reducing service disruption.

Types of Change

Standard change: A low-risk, pre-authorized, repeatable change performed using an established procedure.
Normal change: A change that requires assessment, authorization, planning, and controlled implementation.
Emergency change: A change that must be implemented quickly to resolve a major incident or significant risk.

9. Continual Improvement

Continual Improvement identifies opportunities to improve services, processes, performance, and customer outcomes. Improvement decisions should be supported by measurable evidence such as incident trends, SLA performance, customer feedback, cost, quality, and operational risk.

10. ITSM Metrics

Common ITSM metrics include Mean Time to Restore Service (MTTR), first contact resolution rate, incident volume, repeat incident rate, SLA compliance, backlog, change success rate, and customer satisfaction.

RAG Test Questions and Expected Knowledge

The following questions can be used to validate whether the AI Knowledge Assistant retrieves the correct sections from this document.

Question: What is an incident?

Expected answer: An unplanned interruption to an IT service or reduction in service quality.

Question: What is the objective of Incident Management?

Expected answer: Restore normal service operation quickly and minimize business impact.

Question: What is the difference between impact and urgency?

Expected answer: Impact measures the extent of effect; urgency measures how quickly action is required.

Question: What is a P1 incident?

Expected answer: A critical incident with very high impact and high urgency, such as a major business service outage.

Question: What is a problem?

Expected answer: A cause or potential cause of one or more incidents.

Question: What is the difference between Incident and Problem Management?

Expected answer: Incident Management restores service; Problem Management addresses underlying causes.

Question: What is Root Cause Analysis?

Expected answer: A structured approach for identifying the fundamental cause rather than only treating symptoms.

Question: What is a standard change?

Expected answer: A low-risk, pre-authorized, repeatable change using an established procedure.

Question: What is MTTR?

Expected answer: Mean Time to Restore Service.

Question: What are common ITSM metrics?

Expected answer: MTTR, incident volume, SLA compliance, backlog, change success rate, and customer satisfaction.